Development applications
Engineering submission requirements

We want to streamline and reduce our DA assessment times

Blacktown City is one of the fastest growing local government areas in Australia. We assess more than twice the number of DAs than the average Sydney council. This places our assessment teams under pressure to evaluate DAs in a timely manner.

This document assists applicants to understand the reasons, and the minimum DA submission requirements, for engineering information of proposed developments in Blacktown City.

This document does not provide information on town planning or other DA requirements. Details on the range of information required for a DA is on our website, see links at the end of this document.

We need to ensure our assessment teams gain a reasonable understanding of a DA

Consultants generally spend considerable time preparing DAs and develop a sound understanding of the form and arrangement of a proposed development.

We, by comparison, have limited time in which to gain a reasonable understanding of a development proposal. Therefore, in order to provide a determination and consent to a DA, it is critical that it is concise, complete and accurate; it addresses all applicable development controls and is underpinned by sound engineering calculations. The DA must include technical drawings, reports and computer models (where relevant), setting out the fundamental land use decisions of a proposed development.

Land use decisions affect the form, size and viability of a proposed development, the footprint of buildings, the size of proposed treatment devices, extent of earth works and batters or height of retaining walls, the potential impact of a development on-site and off the site on neighbours and on the environment. An example of a land use decision is the height of a building or fill material required to create a building platform, the volume or footprint of stormwater detention or water quality treatment area or the proposed site coverage.

Land use decisions also include the proposed operational activities and end use that will occur on site; for example, is the proposed development a waste management facility which may have potential transport, air and surface water quality impacts. Land use decisions impact on what we would see constructed and the general arrangement of a site.

All land use decisions must be resolved at DA stage while construction details can be provided at construction certificate stage.

For civil engineers, one specific key land use decision which affects every development relates to ‘how the proposed development will impact on and integrate with adjoining land, utilities and infrastructure’. For example,

> How do the proposed road levels integrate with adjoining road levels?
>
> How do drainage infrastructure levels integrate with drainage infrastructure up or downstream of the proposed development?

Can a proposed bioretention system drain freely to the street or trunk drainage system or does the proposed development need to be raised?

Another critical land use decision relates to the ability of all relevant design vehicles to use the proposed paths of travel both vertically and horizontally. For example, can a commercial vehicle exit from a proposed driveway without travelling into the path of oncoming vehicles.

These are examples of key land-use decisions that must be resolved at DA stage and with sufficient information provided to Council on DA drawings and supporting information to enable Council to see what has been resolved and how the DA addresses these key land use decisions.

We have specific submission requirements for proposed developments

As development coordinator, our role prior to DA submission, is to share information with development proponents on existing or proposed levels, drainage and transport strategies and other relevant plans which would impact a development proposal.

Development proponents are strongly encouraged to view our website, to interrogate the reference documents at the end of this document and to liaise with us at pre-lodgement stage and again throughout the design phase to ensure the most up to date information is used and therefore to understand what development constraints exist.

When submitting a DA, our minimum engineering requirements are set out below. This is not a complete list and is intended only to broadly define typical submission requirements. Site specific requirements may vary from these. Consultants must complete DA submission checklists which provide a complete list of submission requirements.

General arrangement plan	This must show and clearly differentiate existing landforms, built environment and infrastructure (roads, drainage and services). Good drawings clearly show existing features in light grey and proposed works in black.
Catchment plan	This must show contours and catchment extents, overland flow path directions and trunk drainage features. We must be able to quickly understand where stormwater is coming from, how it is to travel through a proposed development site and where it travels to. The catchment plan should show flood extents and flood levels where applicable. We can provide this information on request.
Vertical elevations (long & cross sections)	These must be provided for key elements of a proposal. Key elements are those that affect land use decisions on the site, for example proposed road elevations (long and cross sections) and pipe and channel long sections, detention basins and bioretention systems. Water quality or stormwater detention infrastructure including associated pipework needs to be shown both in plan and in section to clearly demonstrate that the proposal can drain freely and is not influenced by backwater effects. The elevations need to demonstrate the proposed design levels are suitable and that additional filling (and associated retaining walls) is not required.
Impacts on adjoining land	Impacts on adjoining land need to be clearly identified and written permission from adjoining owner's needs to be included in the DA. If a fill batter crosses a site boundary this will need adjoining owner consent.
Computer modelling	Submitted computer models, whether RAFTS, DRAINS and/or MUSIC files MUST match the plans submitted exactly. Frequently they don't and we can't assess the work.

Our reference documents, tools and information

The following reference documents describe our planning and development controls and associated standards and procedures. They include development checklists and water sensitive urban design (WSUD) requirements.

Checklists	What makes an application complete? - Blacktown City
Planning controls and tools	Blacktown Council Mapping including Land use zones, General Planning, Flooding and Bush Fire Blacktown planning controls - Blacktown City including LEP and DCP (2015) North-West-Growth-Centre-planning-controls including SEPP and DCP Planning Controls relating to WSUD including explanations of compliance pathways such as precinct scale water quality offset schemes and fact sheets.
General engineering guidelines	The Engineering Guide for Development & Works Specification Civil
Water sensitive urban design (WSUD) guidelines and tools	WSUD Standard Drawings The Small-Scale Stormwater Quality Model (S3QM) which can be used to: a. initially understand which development controls are applicable to your land b. model any on-site stormwater detention requirements, and c. size on-site water quality treatment and water conservation measures as an alternative to MUSIC. Small Scale Stormwater Quality Assessment Tool (S3QM) WSUD Handbook including MUSIC modelling Requirements Planting and Landscaping Requirements WSUD Signage Requirements Maintenance of WSUD Assets